Section 3 - Dimensionality Reduction Techniques

Motivation:

Before running any algorithms, we needed to prepare the data. Since we are moving to an unsupervised learning approach, we decided to drop the “track_popularity” variable entirely. This allows us to focus purely on the actual acoustic characteristics of the songs, without the noise or unpredictability of commercial success.

Principal Components Analysis

For the Principal Component Analysis (PCA), we selected seven continuous audio features: energy, tempo, danceability, loudness, duration_ms, valence, and liveness. Because these variables are measured on completely different scales, for example, duration is measured in hundreds of thousands of milliseconds, while energy is strictly bounded between 0 and 1, it was necessary to standardise the data.

Following the standardisation, we computed the sample correlation matrix. Then, we extracted the eigenvalues and eigenvectors from this matrix. The eigenvectors serve as our PC loadings, because they define the precise directions of our new orthogonal axes within the high-dimensional space. Meanwhile, the corresponding eigenvalues quantify the magnitude of variance captured along each of these new axes. By performing a matrix multiplication between our standardised dataset and the PC loadings, we derived the actual Principal Components.

The primary objective of this transformation is to reduce the dimensionality of our dataset while preserving as much of the original variance as possible. To determine exactly how many components to retain, we relied on the Scree Plot supported by the Kaiser Criterion. As established by Hair et al. (2019, p.161), the Kaiser rule dictates that we should only retain principal components with an eigenvalue greater than 1, since each original variable contributes exactly 1 unit of variance; a component is therefore only considered “useful” if it explains more variance than a single original variable.

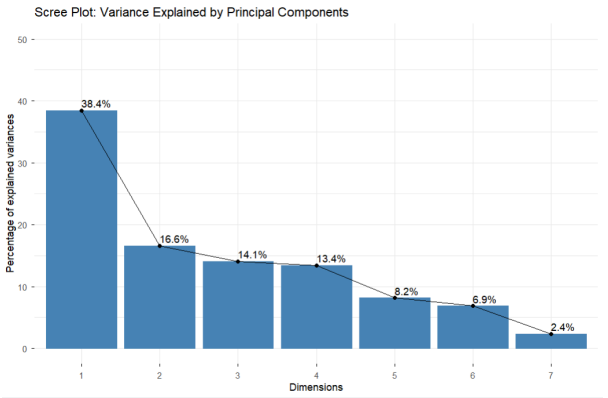


Figure 1: Scree Plot - Variance Explained by Principal Components

The first two dimensions possess eigenvalues well above 1 (PC1 = 2.6902, PC2 = 1.1625), accounting for 38.4% and 16.6% of the variance, respectively. However, applying the strict Kaiser rule would leave out relevant information from tempo and liveness, which are poorly represented in these two components alone. We therefore adopted a Relaxed Kaiser Criterion (retaining eigenvalues > 0.9): PC3 and PC4 present eigenvalues just below 1 (PC3 = 0.9857, PC4 = 0.9415), and the Scree Plot (Figure 1) confirms a clear elbow only after the fourth component, where the curve drops from 13.4% to 8.2% before flattening. Combining both criteria, we retained the first

four principal components, compressing the seven original features into four independent axes that together capture 82.5% of the acoustic variance.

To interpret the underlying meaning of these four retained dimensions, we calculated the correlations between the original audio variables and the derived Principal Components. By squaring these correlations, we obtained the square cosine (cos^2), which measures the representation quality of each variable on the factor maps. In the visualisations below, variables are depicted as vectors. The colour gradient of these vectors indicates their contribution score, which represents a variable's combined contribution to the specific two-dimensional plane being plotted (e.g., PC1 and PC2 together). While a warm colour indicates a strong overall influence on the 2D space, the precise angle of the vector explains its specific effect: variables whose vectors run parallel to and extend close to the edge of a given axis heavily define that specific Principal Component.

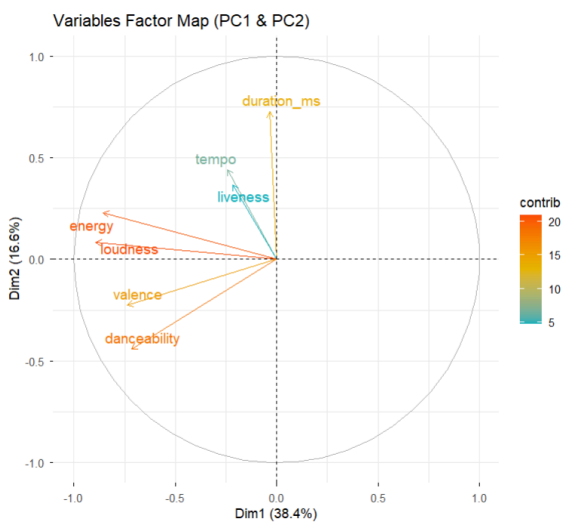


Figure 2: Variables Factor Map (PC1 & PC2)

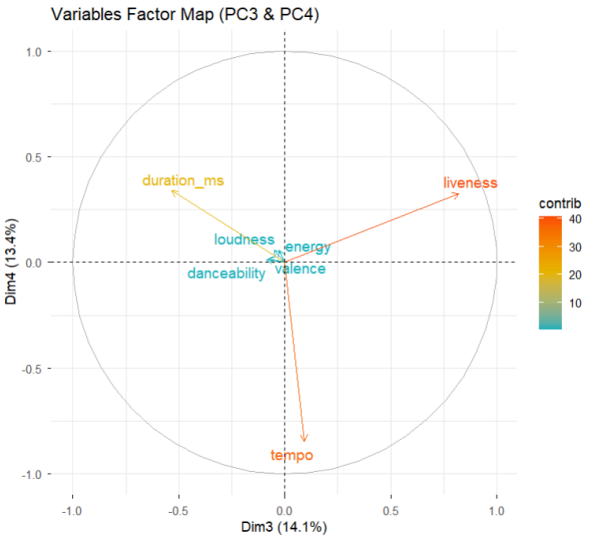


Figure 3: Variables Factor Map (PC3 & PC4)

The Factor Map shown on the left (Figure 2) shows that PC1 (38.4%) is strongly driven by energy and loudness. Both vectors point firmly to the left side of the axis (demonstrating a negative loading on PC1), but because they point in the exact same direction, it confirms a strong positive correlation between them: as a track's physical loudness increases, its perceived energy tends to rise simultaneously. PC1, therefore, represents the acoustic intensity of a track, with lower scores corresponding to higher intensity. PC2 (16.6%) is almost entirely dominated by duration_ms, which points directly upward and forms a near-perfect 90º angle with energy and loudness, confirming that track length is independent of acoustic intensity. Conversely, danceability and valence point towards the lower quadrants, indicating a mild negative correlation with duration: longer tracks tend to be less danceable and exhibit lower valence. Hence, PC2 contrasts long compositions with highly danceable tracks.

The Factor Map on the right (Figure 3) reveals a different set of acoustic dynamics. Variables such as energy, loudness, valence, and danceability show significantly reduced contributions (cooler blue colours), consistent with the mathematical orthogonality of the components. PC3 (14.1%) is defined predominantly by liveness, which stretches far to the right, but it also exhibits a strong negative correlation with duration_ms (-0.534). Together, this axis captures a specific recording context: separating shorter, highly "live" or ambient recordings from longer, heavily studio-produced

tracks. PC4 (13.4%) is heavily influenced by tempo, pointing straight downward, and represents the speed or pacing of the music, isolating fast-paced tracks from slower ones regardless of loudness or energy.

In this last step, we project a categorical variable onto the final PCA plot to help interpret the results. Our original “playlist_genre” variable had 35 different categories, which would have made the final graph cluttered and unreadable. To fix this, we created a new variable called “macro_genre”, grouping the 35 original genres into 6 broader categories based on their musical style: Electronic & Lofi, Pop, Hip-Hop & Afro, Rock & Alternative, Latin & Global, and Acoustic & Classical.

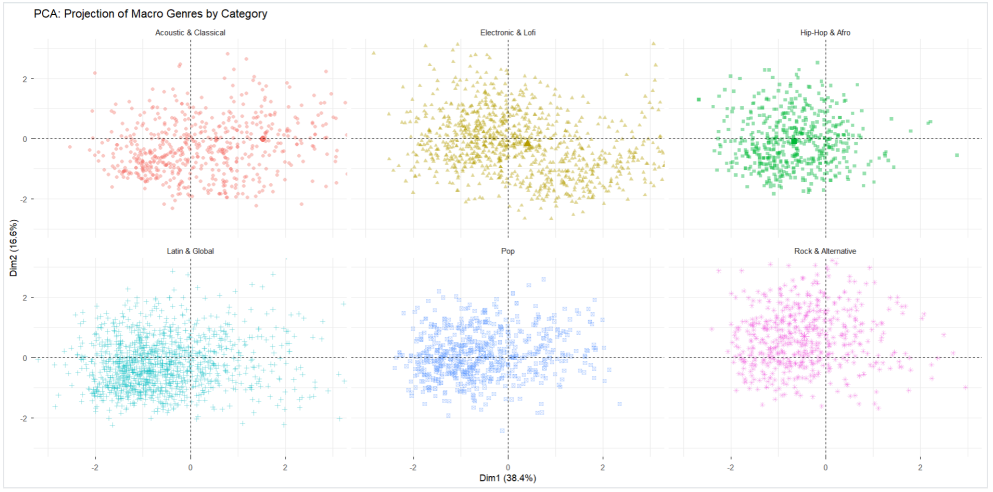


Figure 4: PCA Projection of Macro Genres by Category (PC1 & PC2)

The projection of macro-genres onto PC1 and PC2 (Figure 4) vividly illustrates the high within-group acoustic variance of commercial music. As expected, none of the six genres forms a tight, well-separated cluster. However, distinct directional pulls do emerge. Hip-Hop & Afro tracks shift toward the negative end of PC1, aligning with higher acoustic intensity and energy, while Acoustic & Classical predominantly occupies the opposing positive region, consistent with its quieter, more acoustic nature. Electronic & Lofi displays the widest dispersion across the entire plane; this reflects the massive stylistic heterogeneity introduced by grouping vastly different subgenres (such as ambient study beats and high-energy club music) under a single label, even if led by a synthetic music criterion. Additionally, Latin & Global shows a slight downward shift into the negative region of PC2, suggesting a tendency toward shorter, highly danceable tracks.

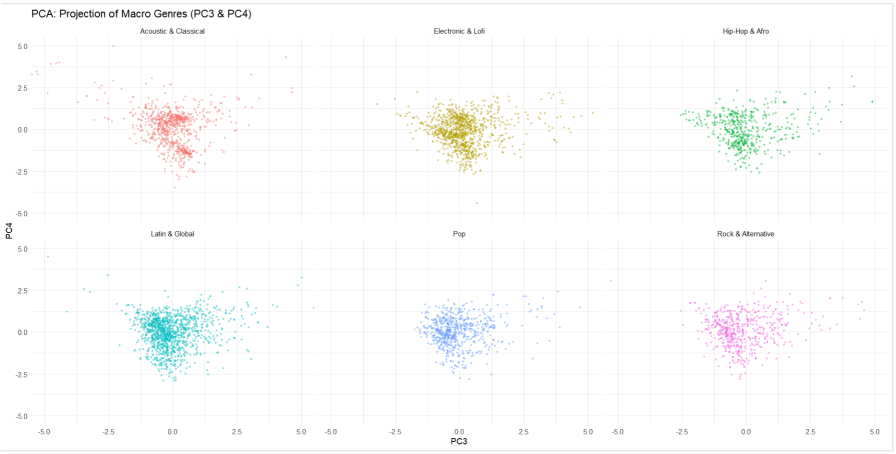


Figure 5: PCA Projection of Macro Genres by Category (PC3 & PC4)

The projection onto PC3 and PC4 (Figure 5) adds further information regarding recording context and pacing. On the PC3 axis, which contrasts liveness with track duration, Acoustic & Classical exhibits a shift toward the positive end, reflecting a higher proportion of live, unedited, or ambient recordings. In contrast, Pop and Electronic & Lofi cluster more centrally and negatively, consistent with tightly controlled, longer, studio-produced output. On the tempo axis (PC4), Acoustic & Classical stretches significantly into the positive upper region, aligning with the characteristically slower tempos of ambient and classical music. Meanwhile, Rock & Alternative and Pop display considerable, messy overlap across both axes. This massive visual overlap across all four principal components reinforces a critical conclusion: culturally assigned genre labels are acoustically inadequate, necessitating an unsupervised clustering approach to find true musical topologies.

The broad overlap across all genres is consistent with Fabbri's (1981) argument that musical genres are defined primarily by social conventions rather than intrinsic sonic properties. This overlap is particularly evident between categories such as Rock & Alternative and Pop, whose point clouds share considerable common ground in the PCA projection. This is further illustrated by Madoery's (2024) model of Rock as a genre existing simultaneously in a broad and a restricted sense, where stylistic territories overlap rather than fit within each other; this is visually confirmed by the massive dispersion in the PCA projection.

Taken together, these results suggest that the macro_genre variable, while useful as an organisational tool, reflects imposed cultural categories more than genuine acoustic discontinuities. Because human-assigned categories are acoustically insufficient, this provides a strong empirical justification for applying unsupervised clustering algorithms in the next section to discover the "true" acoustic groupings of the dataset.

Correspondence Analysis

For the Correspondence Analysis (CA), we were required to analyse two categorical variables with at least three categories each. We selected our engineered macro_genre variable (6 categories) as the first dimension. For the second variable, we converted the continuous tempo variable into a new categorical variable called tempo_category. Instead of splitting the Beats Per Minute (BPM) randomly, we discretised the data into three musically established tiers: Slow (< 100 BPM), Medium (100 - 130 BPM), and Fast (≥ 130 BPM).

Categorising the data in this manner allowed us to construct a robust 6x3 contingency table (Figure 6). This specific discretisation strategy was essential to eliminate the risk of data sparsity (i.e., avoiding cells with expected frequencies below 5), ensuring our data satisfied the strict mathematical assumptions required for a valid Chi-Square test of independence.

Table: Observed Contingency Table: Macro Genres vs. Tempo Categories

	Slow	Medium	Fast
Acoustic & Classical	348	226	248
Electronic & Lofi	274	454	290
Hip-Hop & Afro	138	190	205
Latin & Global	359	462	379
Pop	138	327	180
Rock & Alternative	126	237	205

Figure 6: Observed contingency Table (macro_genre x tempo_category)

Correspondence Analysis is only mathematically meaningful if there is a statistically significant dependency between the categorical variables being analysed. If the genres and tempos were independent, the resulting dimensional space would capture nothing but random noise.

To formally justify the application of CA, we evaluated the hypothesis of independence by comparing our observed contingency table against the expected frequencies calculated under a model of pure independence.

Table: Expected Frequencies under the Hypothesis of Independence

	Slow	Medium	Fast
Acoustic & Classical	237.5	325.6	258.8
Electronic & Lofi	294.2	403.3	320.5
Hip-Hop & Afro	154.0	211.2	167.8
Latin & Global	346.8	475.4	377.9
Pop	186.4	255.5	203.1
Rock & Alternative	164.1	225.0	178.8

Figure 7: Expected frequencies under the Hypothesis of Independence

A visual comparison between the observed and expected tables (Figure 7) reveals important discrepancies, indicating a strong structural relationship. To quantify this, we performed a Chi-Square test of independence. The test returned a Chi-Square statistic of 154.32 with 10 degrees of freedom, yielding a p-value of 2.2e-16. Because this p-value falls far below the standard 0.05 threshold, we decisively rejected the null hypothesis of independence. This result provides empirical proof of a highly significant association between a track's macro-genre and its tempo profile, validating our decision to proceed with the Correspondence Analysis.

To visualise how the genres deviate from the expected independent profiles, we executed the Correspondence Analysis and plotted the resulting dimensions on a Symmetric Biplot.

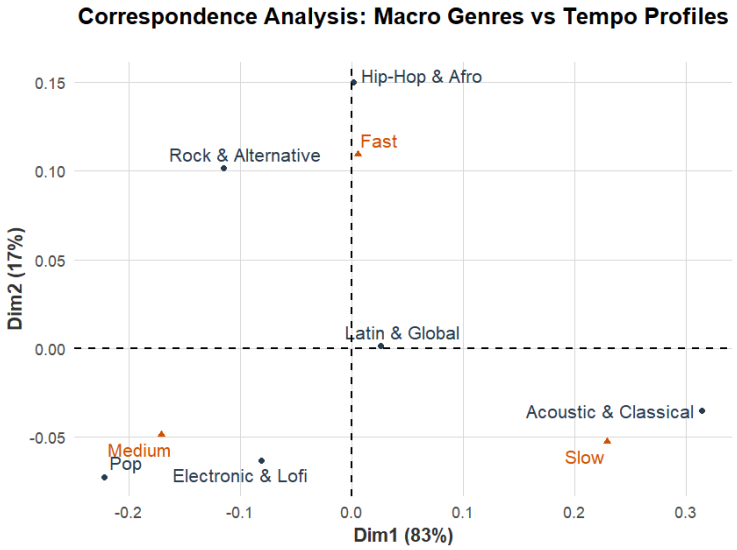


Figure 8: Symmetric Biplot - Correspondence Analysis (macro_genre x tempo_category)

As shown in Figure 8, the first dimension (Dim 1) captured the vast majority of the variance (83%), acting as the primary axis of differentiation, while the second dimension (Dim 2) captured the remaining 17%. When analysing a symmetric biplot, row profiles (Genres, in main colour) and column profiles (Tempos, in red) are projected into different mathematical spaces. Therefore, rather than interpreting the direct Euclidean distance between a genre and a tempo, we interpret the structural associations by looking at angles from the origin and the overall distance from the centre of gravity (the average dataset profile).

- **Acoustic & Classical vs. Slow Tempos:** The biplot reveals a highly acute angle originating from the centre toward both "Acoustic & Classical" and the "Slow" tempo

category on the right side of the plot (positive Dim 1). Furthermore, Acoustic & Classical is positioned very far from the origin. This geometric alignment mathematically proves that this genre has a highly distinct profile that heavily over-indexes in slow tempos (< 100 BPM) compared to the dataset average.

- **Electronic & Lofi and Pop vs. Medium Tempos:** A very strong structural association is visible between the "Electronic & Lofi" genre and the "Medium" tempo point. Both vectors point firmly into the bottom-left quadrant, indicating that electronic tracks in our dataset are fundamentally driven by medium tempos (100 - 130 BPM), and so is Pop, even more extremely.
- **Hip-Hop & Afro vs. Fast Tempos:** Similarly, "Hip-Hop & Afro" is extremely aligned on the Dim2-axis alongside the "Fast" tempo marker. This indicates a strong positive association, demonstrating that this genre features a significantly higher proportion of tracks exceeding 130 BPM. Rock & Alternative follows the same concept and relation with the Fast marker, but deeper into the left quadrant, weakening this association.
- **The "Average" Profile:** Conversely, "Latin & Global" is clustered tightly around the origin (0,0). This central positioning indicates that this genre has a highly generalised tempo profile; It does not deviate significantly from the overall average of the entire dataset, drawing somewhat equally from slow, medium, and fast tracks.

The Correspondence Analysis successfully mapped the structural dependencies between cultural macro-genres and pacing. It mathematically demonstrated that genres exist on a spectrum of structural rigidity: some possess highly specific, rigid tempo signatures, while more commercial categories are completely generalised, lacking any distinct pacing identity. This varying degree of structural rigidity perfectly reinforces the high within-group variance observed in the PCA. Because culturally assigned genres are so acoustically inconsistent, unsupervised clustering presumably remains the most robust and necessary method to discover the true musical topologies of the dataset.

Clustering

In this stage of the study, we transition from dimensionality reduction to unsupervised learning. For the following cluster analysis, we do not use the raw variables from the original dataset. Instead, we use the coordinates of the tracks derived from the first four Principal Components (PCs). There are three fundamental reasons for this approach:

1. **Orthogonality (Non-correlation)**: K-means requires variables to be independent. In our original data, variables like energy and loudness are highly correlated. Using them together would result in “double-counting” their influence and biasing the result. PCA provides us with uncorrelated (orthogonal) axes.
2. **Noise Reduction**: By taking into account only the first four components, which explain 82.57% of the total variance, we effectively filter out the “noise”.
3. **Standardisation**: The PCA scores are already centred and scaled, ensuring that variables with larger units, like duration_ms or tempo, do not dominate the distance calculations over variables with smaller scales, like energy.

A critical step in K-means is the a priori definition of K, the number of clusters. To avoid an arbitrary selection, we implemented the Elbow Method, evaluating the Total Within-cluster Sum of Squares (WSS) for K values ranging from 1 to 10.

The WSS measures the compactness of the clusters; as K increases, the distance between songs and their respective cluster centres (centroids) naturally decreases, causing the WSS to drop. As shown in Figure 9, we look for the “Elbow”, the specific point of inflexion where the marginal gain in

information begins to level off. In our analysis, the bend occurs clearly at K = 3, suggesting that a three-group partition provides the most stable explanation of our data.

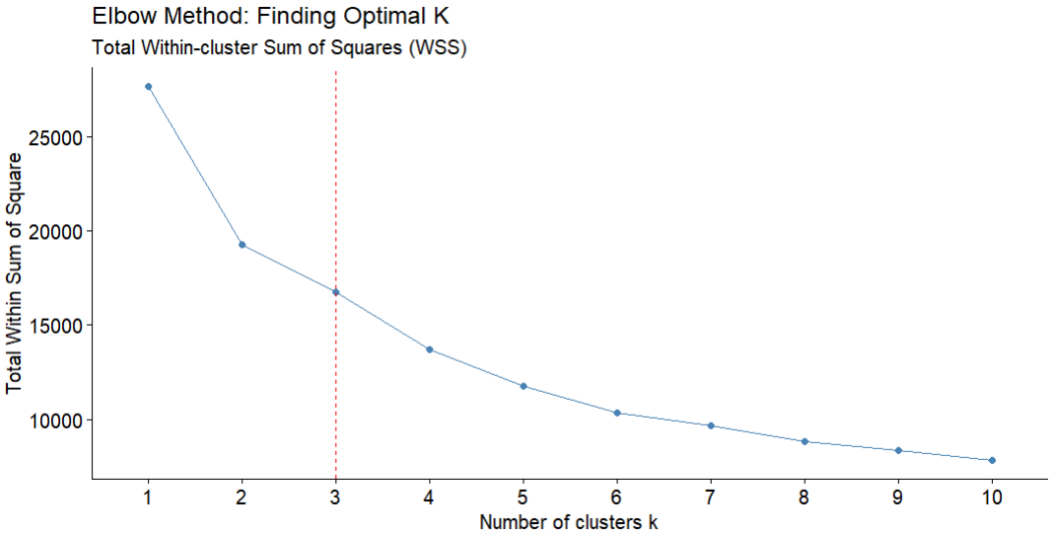


Figure 9: Elbow Plot for determining the optimal number of clusters (K).

Once we decided on three groups (K = 3), we executed the K-means algorithm using Euclidean distance to measure similarity in the 4D space. To ensure the reliability of our results, we utilised 25 random initialisations. Since K-means starts by randomly assigning initial centroids, it can occasionally get stuck. By running the process 25 times, we ensure the results are stable and that the groups we obtain are the best possible mathematical version rather than a random result. The final distribution of these clusters is visualised in Figure 10.

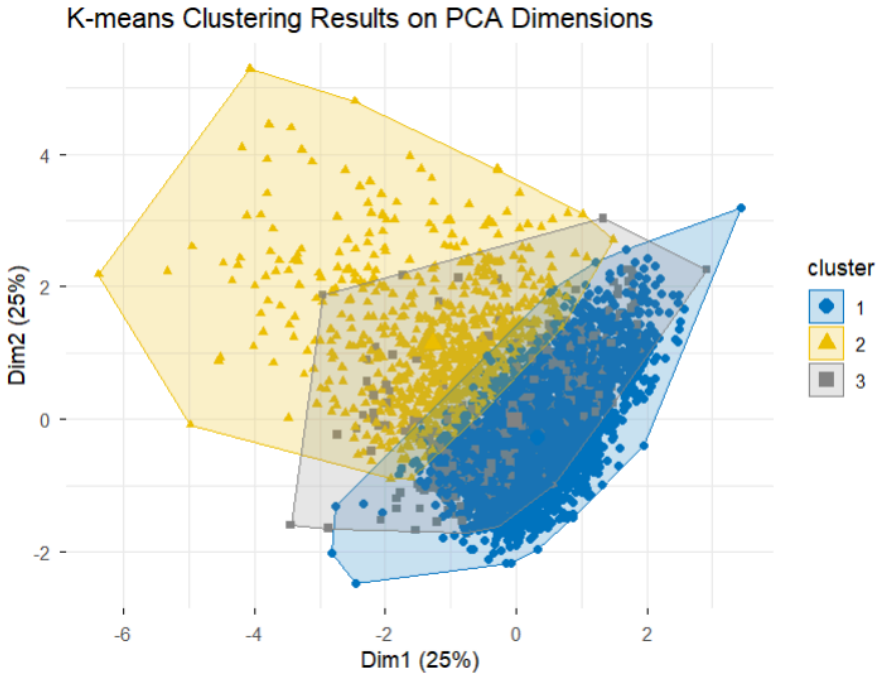


Figure 10: K-means clustering results (K = 3) projected onto the first two Principal Components (PC1 and PC2).

To understand the practical meaning of these three groups, we analysed the mean acoustic values for each cluster. The algorithm reveals three distinct musical types within our dataset based on their physical and emotional characteristics:

1. **Cluster 1: The "Danceable Mainstream" Group**

The cluster is primarily defined by its rhythmic appeal, showing the highest danceability (0.683) and valence (0.543) in the dataset. These tracks are characterised by a positive mood and a high rhythmic consistency. This cluster captures the essence of commercial genres such as Pop, Dance and Urban music types, where high engagement is the primary objective.

2. **Cluster 2: The "High - Intensive" Group**

The second cluster represents the most powerful segment, showing the highest levels of energy (0.731) and loudness (-6.33 dB). Another defining distinction of this group is its liveness score (0.391), which is nearly triple the average of the other clusters. This suggests that the group isolates the "heavier" side of music, such as Rock, Metal or Punk.

3. **The "Atmospheric and Quiet" Group**

Lastly, the third cluster acts as a significant acoustic outlier, characterised by extremely low energy (0.174) and loudness (-21.64 dB). With the lowest valence (0.163) and a slower average tempo (104.5 BPM), this group clearly isolates Acoustic, Classical and Ambient music.

Our analysis demonstrates that the K-means algorithm is capable of grouping music based on its properties without requiring prior knowledge of its genre. In other words, the mathematical results align with our own human perception of what we refer to as 'calm' or 'intense' music. By reducing the data to its core Principal Components, we have revealed a natural organisation of music where the math basically confirms what we feel when we listen to different styles.

Bibliography:

- *Hair, J.F., Black, W.C., Babin, B.J., & Anderson, R.E. (2019). Multivariate Data Analysis (8th ed., p. 161). Cengage Learning EMEA.* *https://eli.johogo.com/Class/CCU/SEM/_Multivariate%20Data%20Analysis_Hair.pdf*

- *Fabbri, F. (1981). A theory of musical genres: Two applications. In D. Horn & P. Tagg (Eds.), Popular Music Perspectives (pp. 52–81). International Association for the Study of Popular Music. https://www.tagg.org/xpdfs/ffabbri81a.pdf*

- *Madoery, D. (2024). El análisis musical genérico/estilístico. Revista del Instituto Superior de Música, 25, e0054.*

  https://bibliotecavirtual.unl.edu.ar/publicaciones/index.php/ISM/es/article/view/13710/19082